

AAEC 5301: Applied Mathematical Economics

Week 1, Lecture 1

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Today

1. Course logistics: content, grading, expectations.
2. Who are we?
3. Chapter 1: what mathematical economics is, and what it is not.
4. Section 2.1: the ingredients of a mathematical model.
5. Section 2.2: the real-number system.
6. Section 2.3: sets and set operations.

We finish Chapter 2 on next lecture.

Contact and meeting times

Instructor	Dr. Ujjwol Paudel
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Office hours	Tue and Thu, 2:00–2:50 PM, or by appointment
Class	Tue and Thu, 3:30–4:50 PM, Ag. Sciences 202

Who am I?

- Assistant Prof. at Applied and Ag Econ Department
- PhD in Ag Econ (Arizona State U.)
- MS in Mathematics (ETH Zürich, CH)
- BS in Mathematics (Jacobs University Bremen, DE)
- Interests: industrial organization economics, operations management
- Other interests: hiking, doing nothing, taking pictures, ...

Some pictures



Some pictures



Some pictures



Your turn:

In roughly 1 min:

1. Your name, undergrad / masters education.
2. Any/what math courses in earlier education?
3. One thing you do that has nothing to do with economics.

Item 2 is important; I will try to structure the course around your background.

What this course is for

We build the mathematical foundation needed in rest of your PhD sequence (micro theory, econometrics):

- matrix algebra and linear systems;
- univariate and multivariate calculus;
- static and comparative-static analysis;
- constrained and unconstrained optimization;
- an introduction to integral calculus.

Text and materials

Required: Chiang and Wainwright (2005), *Fundamental Methods of Mathematical Economics*, 4th ed. We cover Chapters 1–14.

Slides. I post them, but they will not be detailed and are not a substitute for reading the chapter. Book covers examples in great detail.

Software. None required. Maybe some Python/R to check some answers but entirely optional. All exams are handwritten.

L^AT_EX. Suggest you type problem set solutions in L^AT_EX. This is valuable as you will benefit from writing all your future research papers with that tool.

Grading

Problem sets	20%
Quizzes	15%
Midterm I (Week 7, Chapters 1–7)	20%
Midterm II (Week 12, Chapters 8–11)	20%
Final exam (cumulative)	25%

A: 90–100 B: 80–89 C: 70–79 D: 60–69 F: below 60.

Quizzes, problem sets, and make-ups

Quizzes. Most weeks start with a short quiz on last week's material and this week's reading. Mostly multiple choice, occasionally a short answer.

Problem sets. Will target 8 of them. Work them alone first and subsequent discussion with a classmate is fine.

Make-ups. Missed exams are excused only for reasons like university business, documented illness or emergency, or a religious holy day. *An excused midterm transfers its weight to the final, so the final becomes 45%.*

Generative AI policy

Permitted, as **assistance**:

- explaining a concept you did not follow in lecture;
- generating extra practice problems;
- checking an algebraic step you already worked out;
- verifying a derivative, Hessian, or integral you computed by hand.

Not permitted, as **authorship**:

- uploading a problem set and transcribing the output.

Even in the permitted uses: do not outsource the thinking, and verify what the model tells you. These tools are confidently wrong often enough.

One piece of advice

This is a PhD course, so it is no longer about grades. The output you get from this course is directly proportional to the input you give.

Am always open to meeting you for any questions related to this course or academics in general.

Chapter 1

The Nature of Mathematical Economics

What mathematical economics is

It is not a field, the way public finance or trade is a field. It is an **approach** to economic analysis.

The economist states the problem in mathematical symbols and uses known mathematical theorems to reason toward a conclusion.

The subject matter can be anything: micro theory, macro, public finance, labor, political economy, urban, agriculture, supply chain management, marketing, etc.

In the broadest sense, we reserve the term 'mathematical economics' for techniques past simple geometry: matrix algebra, calculus, differential and difference equations.

Mathematical versus literary economics

Any theoretical analysis, mathematical or not, does the same thing: it derives *conclusions* from *assumptions* by a process of reasoning.

Two differences:

1. Assumptions and conclusions are stated in symbols and equations rather than words and sentences.
2. *Proofs* of mathematical *theorems* replace literary logic in the reasoning step.

Symbols and words are equivalents, and we define our symbols in words. But symbols are more convenient in deductive reasoning, and more concise and precise.

Things that are not proofs

- **Proof by example.** It works in one instance. It probably works in all instances.
- **Proof by authority.** My teacher told me.
- **Proof by intimidation.** It is obvious that. . .
- **Proof by exhaustion.** Already has six pages of algebra, so reader will be tired and exhausted to check the details.
- **Proof by notation.** Enough subscripts that nobody will check.
- **Proof by deadline.** It is due at five.

Only the first one is a real temptation in this course, and it has a respectable name: *verification*. It is useful but just not a proof.



Four advantages of mathematical proofs

1. The language is more concise and precise.
2. There is a large stock of existing theorems available to us.
3. It forces every assumption into the open. Theorems are stated “if–then”; to use the “then,” you must first verify the “if.” This is the protection against smuggling in an assumption you did not know you had made.
4. It handles the general n -variable case.

Point (3) is the one that matters most for your research. Most bad economics is not bad algebra. It is an unstated assumption doing quiet work.

Why not just draw the graph?

Geometry is visual, which is a real advantage, and we will use it constantly.

But it has a **dimensional limitation**. Indifference-curve diagrams assume two goods, not because two is realistic, but because three dimensions are hard to draw and four are nearly impossible.

For 3, 4, or n goods there is no picture. There are only equations. That alone is reason enough to go past visual/geometry and use formal language of mathematics.

“But the model is unrealistic”

This is not a valid criticism, and it is not even specific to mathematics.

Theory is *by construction* an abstraction. It singles out the essential factors so we can study the core of a problem without the complications of the actual world. “Theory lacks realism” is therefore a truism, not an objection.

Example: A subway map gets the distances wrong and leaves out the streets. Put them back and it's accurate, and much harder to find your train connection. The distortion is the point.

The useful question is never “is this realistic?” It is “which assumption is doing the work, and does the conclusion survive without it?”

“All models are wrong, but some are useful.” - George Box.

Mathematical economics versus econometrics

Mathematical economics

Applies mathematics to the theoretical side of economic analysis

Deductive: from assumptions to conclusions

Little concern with measurement error

Econometrics

Measures economic data: estimation and hypothesis testing

Inductive: from observations to inference

Measurement error is the central problem

They are complements. Theory must be tested against data before it is trusted; empirical work needs theory to say which regression is worth running.

Concept check 1

Concept check

Which of these is a question for mathematical economics, and which for econometrics?

1. Does a profit-maximizing monopolist always set price above marginal cost?
2. By how much did beef prices rise after the 2020 packing-plant closures?
3. If a per-unit tax is imposed, does the equilibrium price rise by less than the tax?
4. Do farms that adopt precision agriculture have higher yields?

Concept check 1

Concept check

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Answer. 1 and 3 are deductive: they ask what follows from assumptions. Math Econ.
2 and 4 are inductive: they ask what the data show. Econometrics.
The distinction is in the question and not the topic.

Section 2.1

Ingredients of a Mathematical Model

What a model is

The real economy is too complex to understand all at once, and not all of its relationships matter for the question at hand.

So we pick out the factors and relationships that appear primary for our problem and attend to those alone. That deliberately simplified framework is a **mathematical model**.

A mathematical model is a set of equations describing the structure of that framework. The equations give mathematical form to the assumptions; the mathematical operations then deliver conclusions that follow logically from them.

Variables, constants, parameters

A **variable** is something whose magnitude can change: price P , profit π , revenue R , cost C , national income Y . Writing $P = 3$ freezes a variable at a value.

A **constant** does not change. Joined to a variable it is that variable's *coefficient*, as in $7P$.

A **parameter** is a symbolic constant, as in aP . It stands for a given constant, but since we never assign it a number it can take almost any value. Chiang calls it “a constant that is variable.”

Why bother? Solve a model with a in it once, and you have the answer for every value of a . Solve it with 7 in it, and you have the answer for 7.

Endogenous and exogenous

Endogenous variables are the ones the model solves for: the market-clearing price, the profit-maximizing output.

Exogenous variables are determined by forces outside the model. Their magnitudes are accepted as given data.

Parameters and exogenous variables are different things, but both are taken as given when we solve. That resemblance is why many writers loosely call them all parameters.

Convention. Parameters are a, b, c or α, β, γ . Exogenous variables carry a zero subscript: if P is price, P_0 is an exogenously given price.

The label is a modeling choice, not a property

The same variable can be endogenous in one model and exogenous in another.

Wheat price P :

- In a model of the wheat *market*, P is endogenous. Finding it is the point of the exercise.
- In a model of an individual *consumer's* expenditure, P is a datum. The consumer cannot move it. It is exogenous.

Nothing about wheat changed. What changed is where we drew the boundary of the model.

Deciding what is inside and what is outside a model *is* the modeling decision.

Concept check 2

Concept check

A firm chooses how much fertilizer to apply to maximize profit, taking the fertilizer price and the output price as given. Classify each: endogenous, exogenous, or parameter.

1. Fertilizer applied per acre
2. The price of fertilizer
3. The exponent on fertilizer in the production function
4. The firm's profit

Concept check 2

Concept check

A firm chooses how much fertilizer to apply to maximize profit, taking the fertilizer price and the output price as given. Classify each: endogenous, exogenous, or parameter.

1. Fertilizer applied per acre
2. The price of fertilizer
3. The exponent on fertilizer in the production function
4. The firm's profit

Answer. 1 endogenous (it is what the firm chooses). 2 exogenous, written w_0 . 3 parameter. 4 endogenous: profit is determined once the choice is made, even though the firm does not “choose” it directly.

Follow-up: in a model of the *fertilizer market*, what happens to item 2?

Three types of equation

1. **Definitional** equations: an identity between two expressions with the same meaning. Often written with $\equiv$.

$$\pi \equiv R - C$$

This is true by the meaning of the words. It cannot be false.

2. **Behavioral** equations: how one variable responds to changes in another. Consumption responding to income, or cost responding to output.
3. **Conditional** equations: a condition we impose, and then ask what values satisfy it. Quantity demanded equals quantity supplied.

Behavioral equations carry the assumptions

Two cost functions:

$$C = 75 + 10Q$$

$$C = 110 + Q^2$$

- Fixed cost (the value of C at $Q = 0$) is 75 in the first, 110 in the second.
- In the first, each additional unit of output adds a constant 10 to cost.
- In the second, each additional unit adds progressively more.

Same variables, same notation, entirely different technology. It is mainly through the *form* of the behavioral equations that we give mathematical expression to our assumptions.

Conditional equations

Equilibrium conditions state the prerequisite for equilibrium:

$$Q_d = Q_s \quad (\text{quantity demanded} = \text{quantity supplied})$$

$$S = I \quad (\text{intended saving} = \text{intended investment})$$

Optimization conditions state the prerequisite for an optimum:

$$MC = MR$$

These are neither definitional nor behavioral. Nothing forces them to hold. We *impose* them, and then ask what values of the endogenous variables are consistent with them. That is the whole of Chapters 3 through 9.

Concept check 3

Concept check

Classify each as definitional, behavioral, or conditional.

1. $Q_d = 100 - 2P$

2. $Y \equiv C + I + G$

3. $Q_d = Q_s$

4. $AFC \equiv TFC/Q$

Concept check 3

Concept check

Classify each as definitional, behavioral, or conditional.

1. $Q_d = 100 - 2P$

2. $Y \equiv C + I + G$

3. $Q_d = Q_s$

4. $AFC \equiv TFC/Q$

Answer. 1 behavioral — it asserts something about buyers that could be wrong. 2 definitional. 3 conditional. 4 definitional.

The test for a definitional equation: could data ever contradict it? If no, it is a definition.

Putting it together: a market model

$$Q_d = a - bP \quad \text{behavioral} \quad (a, b > 0)$$

$$Q_s = -c + dP \quad \text{behavioral} \quad (c, d > 0)$$

$$Q_d = Q_s \quad \text{conditional}$$

- Endogenous: P, Q_d, Q_s . Three equations, three unknowns.
- Parameters: a, b, c, d . The sign restrictions are where the economics lives — $b > 0$ says demand slopes down.
- Exogenous: nothing yet. Add a per-unit tax t_0 and you have one.

We solve this next week. Today the point is only to see the three ingredients and to know which symbols the model is responsible for producing.

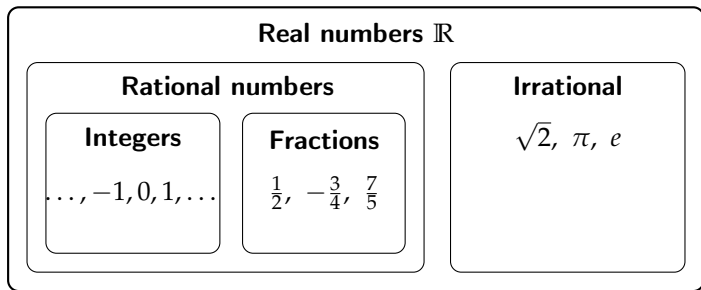
Section 2.2

The Real-Number System

Building up the real numbers

- **Positive integers** $1, 2, 3, \dots$ — the counting numbers.
- **Negative integers** $-1, -2, -3, \dots$, and **zero**, which is neither positive nor negative. Together: the **integers**.
- **Fractions** fill the gaps on the ruler between integers.
- Integers and fractions together are the **rational numbers**: anything expressible as a ratio of two integers. Every integer n qualifies, as $n/1$. Equivalently: a terminating decimal ($\frac{1}{4} = 0.25$) or a repeating one ($\frac{1}{3} = 0.3333\dots$).
- **Irrational numbers** cannot be written as such a ratio: $\sqrt{2} = 1.4142\dots$, $\pi = 3.1415\dots$ — nonrepeating, nonterminating.
- Filling those gaps too gives a continuum: the **real numbers**, denoted $\mathbb{R}$. Displayed on a line, it is the **real line**.

The structure of the real-number system



Read bottom-up and it is a classification of $\mathbb{R}$ into its component sets. Real numbers are all we need in this course.

Concept check 4

Concept check

True or false?

1. $0.272727\ldots$ is irrational.
2. Every integer is a rational number.
3. $\sqrt{9}$ is irrational.
4. There are rational numbers between any two irrational numbers.

Concept check 4

Concept check

True or false?

1. $0.272727\dots$ is irrational.
2. Every integer is a rational number.
3. $\sqrt{9}$ is irrational.
4. There are rational numbers between any two irrational numbers.

Answer. 1 False — it repeats, so it is rational ($3/11$). 2 True, $n = n/1$. 3 False, $\sqrt{9} = 3$; it is the *value* that matters, not the radical sign. 4 True — “density of rationals”; in fact infinitely many.

Section 2.3

The Concept of Sets

Set notation

A **set** is a collection of distinct objects, called its **elements**. Two ways to write one.

By enumeration:

$$S = \{2, 3, 4\}$$

By description:

$$I = \{x \mid x \text{ a positive integer}\}$$

read “ I is the set of all x such that x is a positive integer.”

The vertical bar (or a colon) separates the symbol for a typical element from the description of the elements. Both use braces.

$$J = \{x \mid 2 < x < 5\}$$

Membership, and how big a set can be

Membership is written $\in$:

$$2 \in S, \quad 3 \in S, \quad 8 \notin S, \quad x \in \mathbb{R}.$$

A **finite** set has finitely many elements; $S = \{2, 3, 4\}$. An **infinite** set does not; I and J above.

Finite sets are always **denumerable** (countable): the elements can be put in correspondence with $1, 2, 3, \dots$. Infinite sets may be denumerable (I , the positive integers) or **nondenumerable** (J , an interval of the real line). There is no way to list the reals between 2 and 5 one at a time.

Relationships between two sets

1. **Equal:** identical elements. $\{2, 7, a\} = \{2, a, 7\}$ — order does not matter. One different element and they are not equal.
2. **Subset:** $T \subset S$ if and only if $x \in T$ implies $x \in S$. With $S = \{1, 3, 5, 7, 9\}$ and $T = \{3, 7\}$: $T \subset S$, equivalently $S \supset T$.
3. **Disjoint:** no elements in common. Positive and negative integers.
4. **Overlapping:** some elements shared, some peculiar to each. Neither equal, nor disjoint, nor a subset of the other.

Note the difference: $\in$ relates an *element* to a set; $\subset$ relates a *set* to a set. Writing $3 \subset S$ is a type error.

If $S_1 \subset S_2$ and $S_2 \subset S_1$, then $S_1 = S_2$.

Counting subsets

How many subsets does $S = \{1, 3, 5, 7, 9\}$ have?

Each singleton counts: $\{1\}$, $\{3\}$. So does each pair, triple, quadruple. So does S itself — every element of S is an element of S . A subset that does not contain all of S is a **proper subset**.

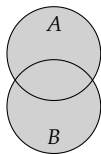
At the other extreme, the **null set** $\emptyset$ (or $\{ \}$) contains no element. It is a subset of every set: if $\emptyset \not\subset S$, then $\emptyset$ would have to contain some $x \notin S$ — but it contains nothing at all.

Total: $2^5 = 32$. In general a set with n elements has 2^n subsets — each element is either in or out, independently.

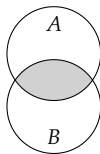
Careful: $\emptyset \neq \{0\}$. The first has no elements; the second has one.

Russell's paradox: In a village, a barber shaves all who don't shave themselves. Who shaves the barber?

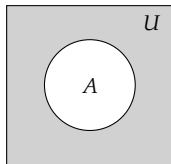
Three operations



$A \cup B$



$A \cap B$



$\tilde{A}$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \quad A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

$$\tilde{A} = \{x \mid x \in U \text{ and } x \notin A\}$$

$\cup$ means “or”, $\cap$ means “and”, the tilde means “not”. Intersection is the more restrictive of the first two.

Examples

Let $A = \{3, 5, 7\}$ and $B = \{2, 3, 4, 8\}$.

$$A \cup B = \{2, 3, 4, 5, 7, 8\}, \quad A \cap B = \{3\}.$$

A and B are neither equal nor disjoint, and neither is a subset of the other.

If $A = \{-3, 6, 10\}$ and $B = \{9, 2, 7, 4\}$, then $A \cap B = \emptyset$. Disjoint sets have an empty intersection.

With universal set $U = \{5, 6, 7, 8, 9\}$ and $A = \{5, 6\}$: $\tilde{A} = \{7, 8, 9\}$. And $\tilde{U} = \emptyset$, since everything under consideration is in U .

From the number-system diagram: integers $\cup$ fractions = rationals; rationals $\cup$ irrationals = $\mathbb{R}$.

Laws of set operations

Commutative:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

Associative:

$$A \cup (B \cup C) = (A \cup B) \cup C, \quad A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Compare $a + b = b + a$, $a \times b = b \times a$, $a \times (b + c) = ab + ac$. The analogy is close, but not perfect: ordinary algebra has only one distributive law, and set algebra has two.

Verifying is not proving

Take $A = \{4,5\}$, $B = \{3,6,7\}$, $C = \{2,3\}$. Check the first distributive law:

$$\text{Left: } A \cup (B \cap C) = \{4,5\} \cup \{3\} = \{3,4,5\}$$

$$\text{Right: } (A \cup B) \cap (A \cup C) = \{3,4,5,6,7\} \cap \{2,3,4,5\} = \{3,4,5\}$$

The two sides agree, so the law is **verified** in this case.

That is not a proof. A single counterexample kills a law, but any number of successful examples does not establish it. To **prove** a law you must show it holds in *all* cases — which means arguing with arbitrary A , B , C and never assigning them values.

On problem sets, read the verb. “Verify” means try an example. “Show” or “prove” means an argument with no specific numbers in it.

Concept check 5

Concept check

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, $B = \{3, 4\}$.

1. Find $A \cap B$.
2. How many subsets does A have?
3. True or false: $\{3\} \in A$.
4. Is $\emptyset \subset B$?

Concept check 5

Concept check

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, $B = \{3, 4\}$.

1. Find $A \cap \tilde{B}$.
2. How many subsets does A have?
3. True or false: $\{3\} \in A$.
4. Is $\emptyset \subset B$?

Answer. 1: $\tilde{B} = \{1, 2, 5, 6\}$, so $A \cap \tilde{B} = \{1, 2\}$. 2: $2^3 = 8$. 3: False — $3 \in A$ and $\{3\} \subset A$, but $\{3\}$ is a set, not an element of A . 4: Yes; the null set is a subset of every set.

Where we are

- Mathematical economics is an approach, not a field. Its main virtue is that it forces assumptions into the open.
- A model is a system of equations: definitional, behavioral, conditional.
- The endogenous/exogenous split is a modeling decision, and it defines what the model is responsible for explaining.
- Parameters buy generality: solve once with a , and you have solved it for every value of a .
- Sets and set operations give us the language for domains, ranges, and constraint sets. This is where they will actually be used, starting in Week 13.

For Thursday

Read: Chiang and Wainwright, Sections 2.4–2.7.

We will cover: ordered pairs and Cartesian products; relations versus functions; domain and range; types of function; functions of several variables; and levels of generality.

Try before Thursday: Exercise 2.3, problems 2, 3, 7, and 9. Problem 9 is the interesting one: it asks how $\emptyset$ can be both a subset of U and the complement of U .

Questions?